

Inference Report – “Image Caption Generator”

1. Project Overview

This project focuses on building an Image Caption Generator using deep learning techniques. The system takes an input image and generates a descriptive caption that explains what is happening in the image. The model is based on a pre-trained BLIP (Bootstrapping Language-Image Pretraining) model from Hugging Face, which was fine-tuned on a subset of the COCO dataset.

The goal of the project is to demonstrate how AI can understand visual data and convert it into meaningful natural language descriptions.

2. Objective

The main objective of this project is to:

- Generate accurate captions for images using an AI model
 - Understand the inference process of image-to-text models
 - Evaluate how well the model performs on unseen images
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3. Technologies and Tools Used

- Python
- Google Colab
- Hugging Face Transformers (BLIP model)
- COCO Dataset (subset of images)
- PyTorch
- GitHub for version control

4. Model and Dataset

The system uses a pre-trained BLIP model, which is designed for vision-language tasks such as image captioning.

- The dataset used is the COCO dataset, but due to storage and runtime limitations, only a subset (~600 images) was used.
- The model was fine-tuned for **2 epochs** because of GPU limitations in Google Colab.
- Despite limited training, the model is still capable of generating meaningful captions.

5. Inference Process

The inference process describes how the model generates captions from input images. The steps are as follows:

1. An image is provided as input to the system
2. The image is processed and converted into features
3. The BLIP model analyzes the image features
4. The model generates a caption based on learned patterns
5. The caption is displayed as output

This process allows the system to convert visual data into human-readable text.

6. Sample Inputs and Outputs

Example 1:

Input Image: A dog running in a field

Generated Caption: “A dog running in the grass”

Example 2:

Input Image: A group of people sitting at a table

Generated Caption: “People sitting together at a dining table”

Example 3:

Input Image: A person riding a bicycle

Generated Caption: “A person riding a bike on the street”

These examples show that the model is able to understand common objects and actions in images.

7. Results and Observations

- The model performs well on simple and common images
 - Captions are generally accurate and meaningful
 - Performance is limited due to fewer training epochs
 - The system successfully demonstrates image-to-text generation
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8. Challenges Faced

During the project, several challenges were encountered:

- **GPU limitations** restricted training to only 2 epochs
- **Large dataset size** caused storage issues in Google Colab
- Initial setup using Kaggle API was complex and time-consuming
- Adjustments were made to use a smaller dataset subset

These challenges required changes in implementation to ensure the system runs efficiently.

9. Conclusion

This project successfully demonstrates the use of AI in generating captions from images. The BLIP model was able to produce meaningful descriptions despite limited training.

In the future, the model can be improved by:

- Training on a larger dataset
- Increasing the number of epochs
- Adding evaluation metrics for better performance analysis

Overall, this project highlights the potential of combining computer vision and natural language processing in real-world applications.